

Peihong Yu

Education

- 2019–2025 **University of Maryland, College Park, Maryland, United States**
Doctor of Philosophy in Computer Science.
Advisor: [Pratap Tokekar](#)
- 2016–2020 **University of Chinese Academy of Sciences, Shanghai, China**
(Associated program with **ShanghaiTech University**)
Master of Science in Engineering, Communication and Information Systems.
Advisors: [Jingyi Yu](#), [Laurent Kneip](#)
- 2012–2016 **Shanghai University, Shanghai, China**
Bachelor of Engineering, Computer Science and Technology
Ranking: 10/285

Employments

- 08/2025–Now **Senior Machine Learning Engineer** at *XMotors.AI*.
- 05-12/2023 **Research Intern, Multi-Agent Reinforcement Learning (PhD)** at *SRI International, CVT*.
- 05-12/2022 **Research Intern, 3D Computer Vision/Machine Learning (PhD)** at *Meta, Reality Lab*.

Publications

* denotes equal contribution.

- NeurIPS'25 **On the Global Optimality of Policy Gradient Methods in General Utility Reinforcement Learning.**
Anas Barakat, Souradip Chakraborty, [Peihong Yu](#), Pratap Tokekar, Amrit Singh Bedi.
The Thirty-ninth Annual Conference on Neural Information Processing Systems, 2025.
- IEEE VIS'25 **VISANATOMY: An SVG Chart Corpus with Fine-Grained Semantic Labels.**
Chen Chen, Hannah K. Bako, [Peihong Yu](#), John Hooker, Jeffrey Joyal, Simon C. Wang, Samuel Kim, Jessica Wu, Aoxue Ding, Lara Sandeep, Alex Chen, Chayanika Sinha, Zhicheng Liu.
IEEE Transactions on Visualization & Computer Graphics, 2025.
- MOO @ RSS'25 **Equitable Optimization Across Preferences: A Maximinimalist Approach to Persistent Monitoring with Preference-based Reinforcement Learning.**
Manav Mishra, [Peihong Yu](#), Sujit PB, Pratap Tokekar.
Workshop on Multi-Objective Optimization and Planning in Robotics at RSS 2025.
- RSS'25 **Sketch-to-Skill: Bootstrapping Robot Learning with Human Drawn Trajectory Sketches.**
[Peihong Yu*](#), Amisha Bhaskar*, Anukriti Singh, Zahiruddin Mahammad, and Pratap Tokekar.
Robotics: Science and Systems, 2025.
- DC @ AAMAS'25 **Learning with Less Effort: Efficient Training and Generalization in (Multi-)Robot Systems**
[Peihong Yu](#).
Doctoral Consortium at International Conference on Autonomous Agents and Multi-Agent Systems, 2025

- AAMAS'25 **TACTIC: Task-Agnostic Contrastive pre-Training for Inter-Agent Communication.**
Peihong Yu, Manav Mishra, Syed Zaidi, and Pratap Tokekar.
International Conference on Autonomous Agents and Multi-Agent Systems, 2025
- TMLR'25 **Beyond Joint Demonstrations: Personalized Expert Guidance for Efficient Multi-Agent Reinforcement Learning.**
Peihong Yu, Manav Mishra, Alec Koppel, Carl Busart, Priya Narayan, Dinesh Manocha, Amrit Singh Bedi, and Pratap Tokekar.
Transactions on Machine Learning Research, 2025.
- OOD @ CoRL'23 **Enhancing Multi-Agent Coordination through Common Operating Picture Integration.**
Peihong Yu, Bhoram Lee, Aswin Raghavan, Supun Samarasekara, Pratap Tokekar, James Zachary Hare.
First Workshop on Out-of-Distribution Generalization in Robotics at CoRL 2023.
- IEEE Access'20 **Accurate Line-Based Relative Pose Estimation with Camera Matrices.**
Peihong Yu, Cen Wang, Zhirui Wang, Jingyi Yu, and Laurent Kneip.
IEEE Access, 2020.
- ICCV'17 **Ray Space Features for Plenoptic Structure-from-Motion.**
Yingliang Zhang, Peihong Yu, Wei Yang, Yuanxi Ma, and Jingyi Yu.
IEEE International Conference on Computer Vision, 2017.
- ICCP'17 **The Light Field 3d Scanner.**
Yingliang Zhang, Zhong Li, Wei Yang, Peihong Yu, Haiting Lin, and Jingyi Yu.
IEEE International Conference on Computational Photography, 2017.

Preprints

* denotes equal contribution.

- Preprint **VARP: Reinforcement Learning from Vision-Language Feedback with Agent Regularized Preferences**
Amisha Bhaskar*, Anukriti Singh*, Peihong Yu, Souradip Chakraborty, Ruthwik Dasyam, Amrit Singh Bedi, and Pratap Tokekar.
Arxiv, 2025.
- Preprint **Insta-RS: Instance-wise Randomized Smoothing for Improved Robustness and Accuracy.**
Chen Chen, Kezhi Kong, Peihong Yu, Juan Luque, Tom Goldstein, Furong Huang.
Arxiv, 2021.

Professional Services

- Reviewer ICRA 2023, ICLR 2025, ICLR 2026, RA-L 2025, T-RO 2025.
- Mentoring [AI4ALL Summer Program](#) at UMD.

Technical Skills

- Languages Python, C++, Matlab
- Libraries PyTorch, Tensorflow, PyMARL, RLlib, Stable Baseline 3, OpenAI Gym
- Other Tools ROS, Gazebo, Unreal Engine